might be a flash drive not connected to the Internet, a hard copy, or a vaulting device. Partial custody combines self-custody and external solution (e.g., Bitgo). Here, a hack on the external provider provides insufficient information to recreate the private key. However, if users lose their private key, the user combined with the external solution can recreate the key. The final option is third-party custody. Many companies that have traditionally focused on custody in centralized finance are now offering solutions in decentralized finance (e.g., Fidelity Digital Assets).

Retail investors generally face two options. The first is self-custody, where users have full control over their keys. This includes a hardware wallet, web wallet (e.g., MetaMask where keys are stored in a browser), desktop wallet, or even a paper wallet. The second is a custodial wallet, in which a third party holds the private keys. Examples are Coinbase and Binance.

The most obvious risk for self-custody is that the private keys are lost or locked. In January 2021, the *New York Times* ran a story about a programmer who used a hardware wallet but forgot the password.[22] The wallet contains $220 million of bitcoin and allows 10 password attempts before all data are destroyed. The programmer has only two tries to go.

Delegated custody also involves risks. For example, if an exchange holds the private keys, it could be hacked and the keys lost. Most exchanges keep the bulk of private keys in "cold storage" (on a drive not connected to the Internet). Nevertheless, there is a long history of exchange attacks, including Mt Gox (2011–2014) 850,000 bitcoin; Bitfloor